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ORIGINAL PAPER

Packing and Covering Triangles in Planar Graphs

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Abstract Tuza conjectured that if a simple graph G does not contain more than k pairwise edge-disjoint triangles, then there exists a set of at most $2k$ edges that meets all triangles in G. It has been shown that this conjecture is true for planar graphs and the bound is sharp. In this paper, we characterize the set of extremal planar graphs.

Keywords Packing and covering · Triangle · Planar graph

Mathematics Subject Classification (2000) 05C70 · 05C35

1 Introduction

We consider finite simple undirected graphs only. For a graph G, we use $V(G)$ and $E(G)$ to denote the vertex set and edge set of G, respectively. A *triangle packing* in G is a set of pairwise edge-disjoint triangles. A *triangle edge cover* in G is a set of

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edges meeting all triangles. We denote by $\nu(G)$ the maximum cardinality of a triangle packing in G, and by $\tau(G)$ the minimum cardinality of a triangle edge cover for G. It is clear that for every graph G we have $\nu(G) \leq \tau(G) \leq 3\nu(G)$.

In [4], Tuza proposed the following conjecture.

Conjecture 1.1 For every graph G, $\tau(G) \leq 2\nu(G)$.

Some partial results have been obtained for certain special classes of graphs. Tuza [5] proved Conjecture 1.1 for planar graphs. Krivelevich [3] extended this result to $K_{3,3}$-free graphs (graphs that do not contain a subdivision of $K_{3,3}$). Haxell and Kohayakawa [2] showed that for tripartite graphs G, $\tau(G) \leq (2-\varepsilon)\nu(G)$, where $\varepsilon > 0.044$. However, the original conjecture of Tuza is still open. The unique non-trivial bound known [1] shows that $\tau(G) \leq \frac{66}{23}\nu(G)$ for every graph G. Note that the inequality $\tau(G) \leq 2\nu(G)$ is sharp for infinitely many planar graphs G, since any planar graph all of whose blocks are isomorphic to K_4 or K_2 reaches this bound. A planar graph G is called *extremal* if it satisfies $\tau(G) = 2\nu(G)$.

In this paper, we shall construct a set of planar graphs $\mathcal{G}$ (which will be defined in Section 2) and prove the following.

Theorem 1.2 *A planar graph G is extremal if and only if $G \in \mathcal{G}$.*

We conclude this section with some notation and terminology. For a graph G, an edge $e \in E(G)$ is *isolated* if e is not contained in any triangle of G. (This definition is useful since isolated edges never affect triangle packings and minimum triangle edge covers, and thus can basically be ignored for our purposes.) An edge e of a triangle T in G is an *own-edge* if T is the unique triangle in G containing e.

We write $A := B$ to rename B as A. For any graph G and any $R \subseteq V(G)$, we use $G[R]$ to denote the subgraph of G induced by R. For any $S \subseteq E(G)$, define $G - S$ to be the subgraph of G with vertex set $V(G)$ and edge set $E(G) - S$. When $S = \{s\}$, we simply write $G - s$ instead of $G - \{s\}$. For any vertex $v \in V(G)$, let $N(v)$ be the set of neighbors of v in G, and let $d(v)$ be the degree of v in G (then $d(v) = |N(v)|$). A vertex v of G is said to be *isolated* if v has degree 0 in G. We use $\Delta(G)$ to denote the maximum degree of G.

2 The Graph Set $\mathcal{G}$

The aim of this section is to construct the set of planar graphs $\mathcal{G}$ and prove some lemmas to be used frequently in later proofs.

The graph set $\mathcal{G}$ is defined as follows. A planar graph $G \in \mathcal{G}$ if and only if there exists a set $\mathcal{S}$ of edge-disjoint K_4's in G such that

(1) each edge in $E(G)$ is either isolated or is an edge of some K_4 in $\mathcal{S}$,
(2) each triangle in G is contained in some K_4 of $\mathcal{S}$.

It is easy to see that any planar graph $G \in \mathcal{G}$ is extremal. Suppose $|\mathcal{S}| = k$, then $\nu(G) = k$ and $\tau(G) = 2k$ (every K_4 in G contributes a triangle for each maximum triangle packing and two edges for each minimum triangle edge cover of G), and hence G is extremal.

The following two observations are immediate from the definition of $\mathcal{G}$.

Proposition 2.1 *Let $G \in \mathcal{G}$, and let $e \in E(G)$ be an edge that is not isolated. Then there exists a minimum triangle edge cover in G containing e.*

Proposition 2.2 *Let $G \in \mathcal{G}$, and let $e, f \in E(G)$ such that neither e nor f is an isolated edge and e, f belong to different K_4's of G. Then there exists a minimum triangle edge cover in G containing both e and f.*

The following lemma was proved by Tuza in [5] when proving Conjecture 1.1 for planar graphs. However, we include the proof for the convenience of the reader.

Lemma 2.3 *Let G be a planar graph. Then G contains a vertex v such that $\Delta(G[N(v)]) \leq 2$.*

Proof Fix a planar embedding of G, we now introduce an algorithm for finding such a vertex v in G.

Let v_0 be an arbitrary vertex in G. If v_0 satisfies the requirement, then we are done. So suppose that v_0 is not an appropriate choice for v. This means that there is a vertex x_0 adjacent to v_0 and having at least three common neighbors with v_0. Then by planarity, one of these common neighbors, say v_1, must be strictly contained inside the triangle with vertex set $\{v_0, x_0, y_0\}$, where $y_0 \neq v_1$ is another common neighbor of v_0 and x_0.

Our next candidate for v is v_1. If v_1 does not satisfy the requirement then, in the same way as before, we can find a vertex v_2 in G, and so on. If G does not contain the desired vertex v, then we can obtain an infinite sequence of vertices $\{v_i : i \geq 0\}$. Furthermore, from the construction of the sequence, we have $v_i \neq v_j$ for all $i, j \geq 0$ and $i \neq j$. But this implies that $|V(G)|$ is infinite, a contradiction. Hence the assertion of the lemma holds. □

Our final lemma in this section deals with extremal planar graphs. A more general version for 3-uniform hypergraphs can be found in [5].

Lemma 2.4 *Let G be an extremal planar graph. Then each triangle in G contains at most one own-edge.*

Proof Let $T := abca$ be a triangle in G. Suppose to the contrary that the lemma is false and assume without loss of generality that both ab and bc are own-edges of G contained in the triangle T.

Consider $H := G - \{ab, bc, ca\}$. Then H is a planar graph, and hence $\tau(H) \leq 2\nu(H)$. Let $\mathcal{M}$ and $\mathcal{C}$ be a maximum triangle packing and a minimum triangle edge cover in H, respectively. Then $\mathcal{M} \cup \{T\}$ is a triangle packing in G, so $\nu(H) \leq \nu(G) - 1$. On the other hand, since both ab and bc are own-edges of G, we see that $\mathcal{C} \cup \{ca\}$ is a triangle edge cover in G. This shows that $\tau(G) \leq \tau(H) + 1$. But then, $\tau(G) \leq \tau(H) + 1 \leq 2\nu(H) + 1 \leq 2(\nu(G) - 1) + 1 < 2\nu(G)$, which contradicts the assumption that G is an extremal planar graph. □

3 Proof of the Main Result

In this section, we prove the main result of this paper.

In fact, we need only to show that any extremal planar graph belongs to $\mathcal{G}$. Suppose for a contradiction that there exists some extremal planar graph which is not in $\mathcal{G}$. Let G be such an extremal planar graph such that $|E(G)|$ is minimum, and subject to this, $|V(G)|$ is minimum. Then by our choice of G, there exists neither an isolated edge nor an isolated vertex in G. Moreover, any extremal planar graph H with $|E(H)| < |E(G)|$ belongs to $\mathcal{G}$.

Let v be a vertex in G described in Lemma 2.3, and let $N := G[N(v)]$. Then by Lemma 2.3, $\Delta(N) \leq 2$. This implies that if N is nonempty, then all the components of N are paths, cycles and isolated vertices.

We claim that N contains no isolated vertex as a component. Otherwise, suppose that w is an isolated vertex in N. Then it is easy to see that vw is an isolated edge in G, contradicting the choice of G.

In the following arguments, we further show that N must be empty.

Lemma 3.1 *N contains no path of odd length as a component.*

Proof Suppose the assertion of the lemma is false and let $P := w_1 w_2 \ldots w_{2k}$ be a path of length $2k-1$ in N (with $k \geq 1$).

Consider $H := G - E(P) - \{vw_i : 1 \leq i \leq 2k\}$ (the edges we remove are the edges shown in Fig. 1a). Then H is planar, and $\tau(H) \leq 2\nu(H)$. Let $\mathcal{M}$ and $\mathcal{C}$ be a maximum triangle packing and a minimum triangle edge cover in H, respectively. Define $T_i := vw_{2i-1}w_{2i}v$ for each $1 \leq i \leq k$ (the thick edges shown in Fig. 1a). Then $\mathcal{M} \cup \{T_i : 1 \leq i \leq k\}$ is a triangle packing in G, and hence $\nu(H) \leq \nu(G) - k$. Furthermore, since $\mathcal{C} \cup E(P)$ (the edges we add to $\mathcal{C}$ are the dashed edges shown in Fig. 1b) is a triangle edge cover in G, we know that $\tau(G) \leq \tau(H) + (2k-1)$. But now, by combining these three inequalities, we have $\tau(G) \leq \tau(H) + (2k-1) \leq 2\nu(H) + (2k-1) \leq 2(\nu(G) - k) + (2k-1) < 2\nu(G)$, which contradicts the assumption that G is extremal. □

Lemma 3.2 *N contains no path of even length as a component.*

Proof Suppose to the contrary that the lemma is false and let $P := w_1 w_2 \ldots w_{2k+1}$ be a path of length $2k$ in N (with $k \geq 1$).

Let $H := G - \{w_i w_{i+1} : 1 \leq i \leq 2k-1\} - \{vw_i : 1 \leq i \leq 2k+1\}$ (note that we do not remove the edge $w_{2k}w_{2k+1}$). Since H is planar, we have $\tau(H) \leq 2\nu(H)$. Let $\mathcal{M}$ and $\mathcal{C}$ be a maximum triangle packing and a minimum triangle edge cover in H, respectively. Define $T_i := vw_{2i-1}w_{2i}v$ for each $1 \leq i \leq k$. Then $\mathcal{M} \cup \{T_i : 1 \leq i \leq k\}$

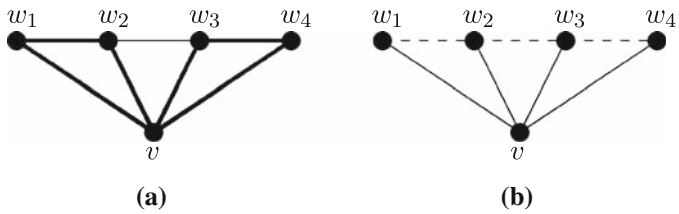

Fig. 1 Lemma 3.1 with $k = 2$

is a triangle packing in G, and $\nu(H) \leq \nu(G) - k$. On the other hand, it is easy to check that $\mathcal{C} \cup E(P)$ is a triangle edge cover in G. This implies that $\tau(G) \leq \tau(H) + 2k$. Now, $\tau(G) \leq \tau(H) + 2k \leq 2\nu(H) + 2k \leq 2(\nu(G) - k) + 2k = 2\nu(G)$. Since G is an extremal planar graph, we see that $\tau(G) = \tau(H) + 2k$, $\tau(H) = 2\nu(H)$, and hence H is extremal. Moreover, because $|E(H)| < |E(G)|$, by our choice of G, we have $H \in \mathcal{G}$.

We claim that $w_{2k}w_{2k+1}$ is not an isolated edge in H. Suppose this is not true and assume that $w_{2k}w_{2k+1}$ is an isolated edge in H. This means that $w_{2k}w_{2k+1}$ is only contained in the triangle $vw_{2k}w_{2k+1}v$ of G. So $w_{2k}w_{2k+1}$ is an own-edge in G. But since vw_{2k+1} is also an own-edge of G contained in the triangle $vw_{2k}w_{2k+1}v$, we know that $vw_{2k}w_{2k+1}v$ contains at least two own-edges of G, which contradicts Lemma 2.4. Hence the claim holds.

Then by Proposition 2.1, there exists a minimum triangle edge cover $\mathcal{C}^*$ in H such that $w_{2k}w_{2k+1} \in \mathcal{C}^*$. But then, $\mathcal{C}^* \cup \{w_iw_{i+1} : 1 \leq i \leq 2k-1\}$ is a triangle edge cover in G of size $\tau(H) + (2k-1)$, contradicting the fact that $\tau(G) = \tau(H) + 2k$. This completes the proof of the lemma. □

Lemma 3.3 *N contains no cycle of even length as a component.*

Proof Suppose for a contradiction that $C := w_1w_2 \ldots w_{2k}w_1$ is a cycle of length $2k$ in N (with $k \geq 2$).

We claim that w_iw_{i+1} is an own-edge of G contained in the triangle $vw_iw_{i+1}v$ for each $1 \leq i \leq 2k$, where $w_{2k+1} := w_1$. Suppose to the contrary that this is not true and assume without loss of generality that w_2w_3 is not an own-edge. Consider $H := (G - w_1w_2) - \{w_iw_{i+1} : 3 \leq i \leq 2k\} - \{vw_i : 1 \leq i \leq 2k\}$ (the edges we remove are the edges shown in Fig. 2a). Then H is planar, and $\tau(H) \leq 2\nu(H)$. Let $\mathcal{M}$ and $\mathcal{C}$ be a maximum triangle packing and a minimum triangle edge cover in H, respectively. Define $T_i := vw_{2i-1}w_{2i}v$ for each $1 \leq i \leq k$ (the thick edges shown in Fig. 2a). Then $\mathcal{M} \cup \{T_i : 1 \leq i \leq k\}$ is a triangle packing in G, and hence $\nu(H) \leq \nu(G) - k$. Further, since $\mathcal{C} \cup E(C)$ (the edges we add to $\mathcal{C}$ are the dashed edges shown in Fig. 2b) is a triangle edge cover in G, we have $\tau(G) \leq \tau(H) + 2k$. Now, by combining these three inequalities, we deduce that $\tau(G) \leq \tau(H) + 2k \leq 2\nu(H) + 2k \leq 2(\nu(G) - k) + 2k = 2\nu(G)$. Since G is extremal, we see that $\tau(G) = \tau(H) + 2k$, $\tau(H) = 2\nu(H)$, and hence H is also an extremal

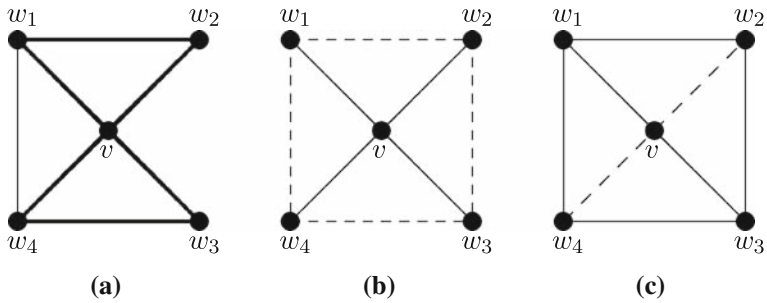

Fig. 2 Lemma 3.3 with $k = 2$

planar graph. Then it follows from $|E(H)| < |E(G)|$ that $H \in \mathcal{G}$. Since we assume that w_2w_3 is not an own-edge of G contained in the triangle vw_2w_3v, w_2w_3 is not an isolated edge in H. Then by Proposition 2.1, there exists a minimum triangle edge cover $\mathcal{C}^*$ in H such that $w_2w_3 \in \mathcal{C}^*$. But now, $\mathcal{C}^* \cup \{w_1w_2\} \cup \{w_iw_{i+1} : 3 \le i \le 2k\}$ is a triangle edge cover in G of size $\tau(H) + (2k-1)$, which contradicts the fact that $\tau(G) = \tau(H) + 2k$. This proves the claim.

We now consider $H' := G - E(C) - \{vw_i : 1 \le i \le 2k\}$. Since H' is planar, $\tau(H') \le 2\nu(H')$. Let $\mathcal{M}'$ and $\mathcal{C}'$ be a maximum triangle packing and a minimum triangle edge cover in H', respectively. Then $\mathcal{M}' \cup \{T_i : 1 \le i \le k\}$ is a triangle packing in G, and $\nu(H') \le \nu(G) - k$. On the other hand, since w_iw_{i+1} is an own-edge of G contained in the triangle $vw_iw_{i+1}v$ for each $1 \le i \le 2k$, it is easy to see that $\mathcal{C}' \cup \{vw_{2i} : 1 \le i \le k\}$ (the edges we add to $\mathcal{C}'$ are the dashed edges shown in Fig. 2c) is a triangle edge cover in G. So $\tau(G) \le \tau(H') + k$. But then, $\tau(G) \le \tau(H') + k \le 2\nu(H') + k \le 2(\nu(G) - k) + k < 2\nu(G)$, contradicting the assumption that G is an extremal planar graph. Hence the assertion of the lemma holds. □

Lemma 3.4 *N contains no cycle of odd length ≥ 5 as a component.*

Proof Suppose for a contradiction that $C := w_1w_2 \ldots w_{2k+1}w_1$ is a cycle of length $2k+1$ in N (with $k \ge 2$).

We claim that w_iw_{i+1} is an own-edge of G contained in the triangle $vw_iw_{i+1}v$ for each $1 \le i \le 2k+1$, where $w_{2k+2} := w_1$. Suppose this is not true and assume without loss of generality that w_2w_3 is not an own-edge. Let $H := (G - w_1w_2) - \{w_iw_{i+1} : 3 \le i \le 2k-1\} - \{vw_i : 1 \le i \le 2k+1\}$ (the edges we remove are the edges shown in Fig. 3a). Since H is planar, $\tau(H) \le 2\nu(H)$. Let $\mathcal{M}$ and $\mathcal{C}$ be a maximum triangle packing and a minimum triangle edge cover in H, respectively. Define $T_i := vw_{2i-1}w_{2i}v$ for each $1 \le i \le k$ (the thick edges shown in Fig. 3a). Then $\mathcal{M} \cup \{T_i : 1 \le i \le k\}$ is a triangle packing in G, and hence $\nu(H) \le \nu(G) - k$. On the other hand, since $\mathcal{C} \cup \{vw_{2k+1}\} \cup \{w_iw_{i+1} : 1 \le i \le 2k-1\}$ (the edges we add to $\mathcal{C}$ are the dashed edges shown in Fig. 3b) is a triangle edge cover in G, we know that $\tau(G) \le \tau(H) + 2k$. By combining these three inequalities, we have $\tau(G) \le \tau(H) + 2k \le 2\nu(H) + 2k \le 2(\nu(G) - k) + 2k = 2\nu(G)$. Since G is an

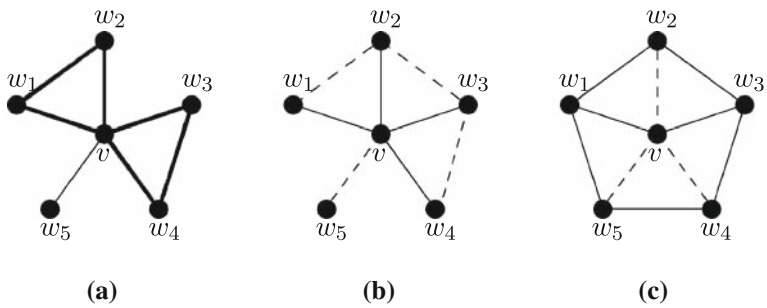

Fig. 3 Lemma 3.4 with $k = 2$

extremal planar graph, we conclude that $\tau(G) = \tau(H) + 2k$, $\tau(H) = 2\nu(H)$, and hence H is also extremal. Moreover, because $|E(H)| < |E(G)|$, by our choice of G, we have $H \in \mathcal{G}$. Since w_2w_3 is not an own-edge of G, w_2w_3 is not an isolated edge in H. Then by Proposition 2.1, there exists a minimum triangle edge cover $\mathcal{C}^*$ in H such that $w_2w_3 \in \mathcal{C}^*$. Now, $\mathcal{C}^* \cup \{w_1w_2, vw_{2k+1}\} \cup \{w_iw_{i+1} : 3 \le i \le 2k-1\}$ is a triangle edge cover in G of size $\tau(H)+(2k-1)$, contradicting the fact that $\tau(G) = \tau(H)+2k$. Hence the claim holds.

Define $H' := G - E(C) - \{vw_i : 1 \le i \le 2k+1\}$. Then H' is planar, and hence $\tau(H') \le 2\nu(H')$. Let $\mathcal{M}'$ and $\mathcal{C}'$ be a maximum triangle packing and a minimum triangle edge cover in H', respectively. Then $\mathcal{M}' \cup \{T_i : 1 \le i \le k\}$ is a triangle packing in G, and $\nu(H') \le \nu(G) - k$. Furthermore, since w_iw_{i+1} is an own-edge of G contained in the triangle $vw_iw_{i+1}v$ for each $1 \le i \le 2k+1$, we see that $\mathcal{C}' \cup \{vw_{2i} : 1 \le i \le k\} \cup \{vw_{2k+1}\}$ (the edges we add to $\mathcal{C}'$ are the dashed edges shown in Fig. 3c) is a triangle edge cover in G. This shows that $\tau(G) \le \tau(H') + (k+1)$. But now, $\tau(G) \le \tau(H') + (k+1) \le 2\nu(H') + (k+1) \le 2(\nu(G) - k) + (k+1) < 2\nu(G)$ (since we assume $k \ge 2$), which contradicts the assumption that G is extremal. This completes the proof of the lemma. □

Lemma 3.5 *N contains no triangle as a component.*

Proof Suppose the assertion of the lemma is false and let $T := abca$ be a triangle in N.

Consider $H := G - \{ab, va, vb, vc\}$. Then H is planar, and $\tau(H) \le 2\nu(H)$. Let $\mathcal{M}$ and $\mathcal{C}$ be a maximum triangle packing and a minimum triangle edge cover in H, respectively. Then $\mathcal{M} \cup \{vabv\}$ is a triangle packing in G, which means that $\nu(H) \le \nu(G) - 1$. On the other hand, it is easy to see that $\mathcal{C} \cup \{ab, vc\}$ is a triangle edge cover in G, and hence $\tau(G) \le \tau(H) + 2$. By combining these three inequalities, we have $\tau(G) \le \tau(H) + 2 \le 2\nu(H) + 2 \le 2(\nu(G) - 1) + 2 = 2\nu(G)$. Since G is extremal, we deduce that $\tau(G) = \tau(H) + 2$, $\tau(H) = 2\nu(H)$, and hence H is an extremal planar graph. Then it follows from $|E(H)| < |E(G)|$ that $H \in \mathcal{G}$.

We claim that at least one edge of $\{bc, ca\}$ is an isolated edge in H. Suppose to the contrary that neither bc nor ca is isolated in H. Since $H \in \mathcal{G}$ and $ab \notin E(H)$, we see that bc and ca are contained in different K_4's of H. Then by Proposition 2.2, there exists a minimum triangle edge cover $\mathcal{C}^*$ in H such that $\{bc, ca\} \subseteq \mathcal{C}^*$. But then, $\mathcal{C}^* \cup \{ab\}$ is a triangle edge cover in G of size $\tau(H) + 1$, which contradicts the fact that $\tau(G) = \tau(H) + 2$. This proves the claim.

Without loss of generality, we may assume that bc is an isolated edge in H. Then bc is only contained in the triangles $\{T, vbcv\}$ of G. Let $H' := G - \{bc, va, vb, vc\}$. By the same argument as above, we can prove that $\tau(G) = \tau(H') + 2$, H' is also extremal, and $H' \in \mathcal{G}$.

We further claim that both ab and ca are isolated in H'. For otherwise, we may assume by symmetry that ab is not an isolated edge in H'. Then by Proposition 2.1, there exists a minimum triangle edge cover $\mathcal{C}'$ in H' such that $ab \in \mathcal{C}'$. But now, $\mathcal{C}' \cup \{vc\}$ is a triangle edge cover in G of size $\tau(H') + 1$ (since bc is only contained in the triangles $\{T, vbcv\}$ of G), contradicting the fact that $\tau(G) = \tau(H') + 2$. Hence the claim holds.

We now consider $J := H' - \{ab, ca\}$. Since $H' \in \mathcal{G}$ and both ab and ca are isolated in H', we have $J \in \mathcal{G}$. It is easy to check that the roles of ab and ca are exactly the same as bc in G, that is, ab is only contained in the triangles $\{T, vabv\}$ and ca is only contained in the triangles $\{T, vcav\}$ of G. Note that $J = G - \{ab, bc, ca, va, vb, vc\}$, we conclude that G is also an extremal planar graph in $\mathcal{G}$, a contradiction. □

We can now prove Theorem 1.2. Since N contains no isolated vertex as a component and by Lemmas 3.1, 3.2, 3.3, 3.4 and 3.5, we know that N must be empty. But then, v is an isolated vertex in G, which contradicts the choice of G. This completes the proof of the theorem. □

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